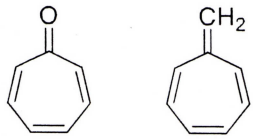
**Organic Cume, 10/26/2018**

1. a) Give the energy levels of the cyclopropenyl cation (qualitative) that you would expect from HMO theory. b) Label the HOMO and LUMO energy levels. . (This question is not asking for a calculation, this should be answered using a <u>qualitative</u> energy diagram of the MO's for the cyclopropenyl cation).

2. Tropone **1** is a thermodynamic stable compound and is easily available for purchase as a starting material for synthesis. However, heptafulvene **2** is relatively unstable, and not commercially available. With this knowledge, speculate why this is the case.



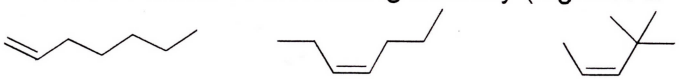

**1** **2**

3. Consider the two $C_7H_{10}$ compounds, **3** and **4**, shown below. a) Which is lower in potential energy and why? b) Calculate the difference in stabilization energy in terms of β.

**3** **4**

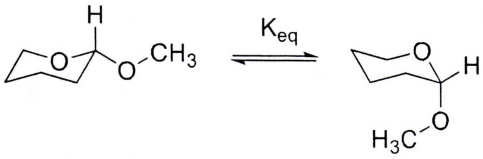
4. Give the preferred conformation for propene.

5. Rank the following alkenes in terms of increasing stability (highest E → lowest E).

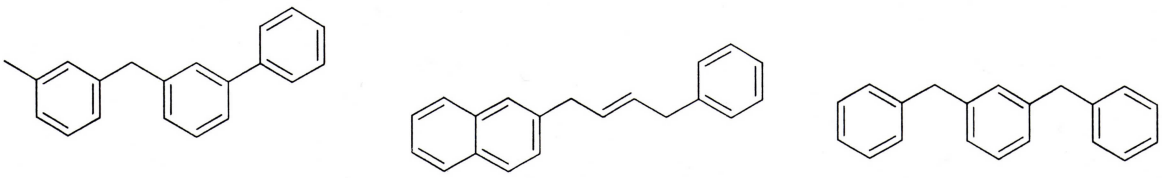


**A** **B** **C**

6. Predict whether the Keq of the following equilibrium is less or greater than 1. Explain.




7. Rank the following $C_{20}H_{18}$ aromatic compounds in terms of increasing stability (highest E → lowest E). Give a reason for your ranking.



**A** **B** **C**